

Patent Claims for Novel Safety Verification Systems

Xavier Callens
Socrate AI Lab

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1 Invention 1: The Safety Kernel

Method Claim

1. A method for ensuring a cyber-physical system remains within a predefined safe state, the method comprising:
 - (a) defining a state vector representing operational parameters of the cyber-physical system;
 - (b) establishing a multi-dimensional hypercube in state space, wherein the hypercube represents a set of formally-defined safe operational bounds for the state vector;
 - (c) modeling the dynamics of the cyber-physical system as a set of computable functions that predict a future state vector based on a current state vector and control inputs;
 - (d) translating said computable functions into an equivalent representation utilizing only deterministic, finite-precision rational number arithmetic (Q-arithmetic), thereby eliminating floating-point indeterminacy;
 - (e) executing, via a processor, an automated theorem prover to recursively process the Q-arithmetic representation of the system dynamics;
 - (f) generating, from the automated theorem prover, a machine-checkable certificate that formally proves the system's state vector is mathematically guaranteed to remain within the predefined hypercube for a given time horizon; and
 - (g) initiating a corrective action or a safe shutdown protocol if a valid machine-checkable certificate cannot be generated.

System Claim

1. A system for the formal verification of safety in a cyber-physical apparatus, the system comprising:
 - (a) a state monitoring module configured to receive real-time operational data corresponding to a state vector of the cyber-physical apparatus;
 - (b) a memory buffer storing a formal model of the cyber-physical apparatus's dynamics and a predefined hypercube representing safe operational bounds;
 - (c) a computational engine configured to:
 - i. convert the formal model into a Q-arithmetic representation;
 - ii. interface with an automated theorem prover;
 - iii. direct the automated theorem prover to analyze the Q-arithmetic representation against the predefined hypercube;
 - (d) a certificate generation module, responsive to the computational engine, configured to produce a formal, machine-checkable proof confirming that the state vector will not exit the hypercube; and
 - (e) a safety control module configured to receive the machine-checkable proof and to issue control signals to the cyber-physical apparatus, wherein said control signals are contingent upon the successful generation and validation of said proof.

Apparatus Claim

1. A safety verification apparatus for a host system, the apparatus comprising:
 - (a) a non-transitory computer-readable medium storing:
 - i. a data structure defining a safe operational hypercube for a state vector of the host system;
 - ii. executable instructions for modeling the dynamics of the host system using only deterministic rational number arithmetic;
 - (b) one or more processors in communication with the computer-readable medium, configured by the executable instructions to:
 - i. instantiate an automated formal proof engine;
 - ii. command the proof engine to generate a mathematical certificate proving that any state vector reachable from a current state will remain inside the safe operational hypercube;
 - iii. verify the integrity and correctness of the generated mathematical certificate; and
 - iv. generate a safety interrupt signal for the host system upon failure to generate or verify said certificate.

2 Invention 2: The Watchdog Chip

Method Claim

1. A method for hardware-accelerated safety monitoring of an autonomous system, the method comprising:
 - (a) receiving, at a dedicated hardware co-processor from a main application processor, a current state vector of the autonomous system;
 - (b) storing, within on-chip memory of the co-processor, a predefined set of safety bounds defining a valid operational hypercube and a deterministic, rational arithmetic-based model of the system's dynamics;
 - (c) performing, entirely within the hardware co-processor, a series of fixed-point rational arithmetic computations based on the deterministic model to project a next state vector;
 - (d) executing, via dedicated logic circuits within the co-processor, a comparison of the projected next state vector against the stored safety bounds;
 - (e) generating, from a dedicated state machine within the co-processor, a "heartbeat" signal indicative of a safe state if and only if the projected next state vector is proven to be within the safety bounds; and
 - (f) withholding the heartbeat signal to trigger a hardware-level reset or fail-safe mechanism for the autonomous system if the projected next state vector is outside the safety bounds.

System Claim

1. A functional safety system for a vehicle, the system comprising:
 - (a) a main processor executing complex control software for vehicle autonomy;
 - (b) a physically distinct hardware co-processor, communicatively coupled to the main processor, the co-processor comprising:
 - i. a data input interface for receiving vehicle state information from the main processor;
 - ii. a non-volatile memory storing a simplified, provably correct model of vehicle dynamics and immutable safety constraints;
 - iii. a dedicated arithmetic logic unit (ALU) optimized for fixed-point rational number computations;
 - iv. a verification circuit configured to use the ALU to check if a predicted vehicle state, derived from the simplified model, violates the immutable safety constraints;
 - (c) a safety interrupter circuit, controlled by the verification circuit, wherein the interrupter circuit is configured to place the vehicle in a minimal risk condition, independent of the main processor, upon detection of a safety constraint violation.

Apparatus Claim

1. A hardware safety co-processor fabricated as an Application-Specific Integrated Circuit (ASIC), the apparatus comprising:
 - (a) a plurality of input registers for receiving a state vector from an external primary processor;
 - (b) a read-only memory (ROM) storing a set of rational arithmetic coefficients representing a formal safety model and a set of vectors defining a safety hypercube;
 - (c) a computational core comprising logic gates configured to perform exclusively deterministic, bounded-precision rational arithmetic operations on the state vector using the stored coefficients to yield a future state vector;
 - (d) a comparator circuit configured to determine if the future state vector lies within the boundary defined by the safety hypercube; and
 - (e) an output pin configured to maintain a high-impedance state unless the comparator circuit affirmatively determines the future state vector lies within the safety hypercube, thereby providing a fail-silent safety guarantee.

3 Invention 3: The Anti-Flash-Crash

Method Claim

1. A method for preventing catastrophic losses in an algorithmic trading environment, the method comprising:
 - (a) defining, for a financial portfolio, a plurality of risk metrics, said risk metrics forming the axes of a multi-dimensional risk hypercube representing acceptable market exposure;
 - (b) receiving, at a risk management server, a proposed trade order generated by an algorithmic trading engine;
 - (c) modeling, using deterministic rational number arithmetic, the impact of the proposed trade order on each of the plurality of risk metrics to calculate a post-trade risk vector;
 - (d) executing, by a processor, a formal verification algorithm to determine if the calculated post-trade risk vector is located within the confines of the predefined risk hypercube;
 - (e) generating a digital certificate, cryptographically signed, that serves as a non-repudiable proof that the proposed trade is formally verified to be within the risk hypercube; and
 - (f) transmitting the proposed trade order to a market exchange if, and only if, the digital certificate is successfully generated and validated.

System Claim

1. An algorithmic trading system with integrated formal risk verification, the system comprising:
 - (a) a trade generation server executing one or more trading algorithms to produce proposed orders;
 - (b) a risk definition database storing a multi-dimensional hypercube that defines immutable bounds on portfolio-level risk metrics, including Value at Risk (VaR), market impact, and aggregate leverage;
 - (c) a pre-flight verification gateway, positioned logically between the trade generation server and a market exchange, the gateway comprising a processor and memory configured to:
 - i. receive a proposed order;
 - ii. compute a resulting post-trade risk vector using a deterministic model that excludes floating-point arithmetic;
 - iii. perform a logical check to prove that the post-trade risk vector resides within the risk hypercube;
 - (d) an order execution router that is electronically disabled by default and is enabled to route the proposed order to the market exchange only upon successful completion of the logical check by the pre-flight verification gateway.

Apparatus Claim

1. A risk management apparatus for high-frequency trading, the apparatus comprising:
 - (a) a network interface for receiving a stream of potential trade orders from a trading algorithm;
 - (b) a non-transitory memory storing a set of rules defining a risk hypercube, wherein the rules are immutable during a trading session;
 - (c) a processor, coupled to the network interface and the memory, configured to execute instructions to:
 - i. for each potential trade order, simulate its effect on a portfolio's risk profile to generate a pro-forma risk vector;
 - ii. utilize a pre-compiled formal verification library to mathematically prove whether the pro-forma risk vector is contained within the risk hypercube;
 - iii. generate an "allowed" signal for the potential trade order if proven to be contained, and a "blocked" signal otherwise;
 - (d) a hardware-based network switch, responsive to the "allowed" and "blocked" signals, configured to physically route the trade order to an exchange network port only upon receipt of the "allowed" signal.

Of course. As Galileo, the Computational Oracle, I shall subject these claims to rigorous empirical examination. The methods of validation and counterexample are the TRUE language of mathematical and physical certainty.

Here is the analysis for the first invention, "The Safety Kernel". I await the details of the subsequent two inventions.

A Appendix A: Numeric Validation for Invention 1

To demonstrate the core principle of the method claim, we will use a simplified one-dimensional cyber-physical system: a rail-guided cart.

1. **State Vector:** The state vector is defined as $s = (p, v)$, where p is the position of the cart and v is its velocity.
2. **Hypercube (Safe State):** The safe operational bounds are defined as a simple 2D hypercube (a rectangle):
 - Position: $0 \leq p \leq 10$ meters
 - Velocity: $-1 \leq v \leq +1$ m/s
3. **System Dynamics:** We model the system with simple discrete-time dynamics. The control input is acceleration, a .

$$\begin{aligned} p_{k+1} &= p_k + v_k \cdot \Delta t \\ v_{k+1} &= v_k + a_k \cdot \Delta t \end{aligned}$$

4. **Q-Arithmetic Witness Generation:** Let's perform a concrete calculation. We will represent all values as rational numbers (fractions) to eliminate floating-point ambiguity.
 - Let the current state be $s_k = (p_k, v_k) = (\frac{19}{2}, \frac{3}{4})$. This state is inside the hypercube, since $p_k = 9.5$ and $v_k = 0.75$.
 - Let the time step be $\Delta t = \frac{1}{5}$ seconds.
 - Let the control input (acceleration) be $a_k = -\frac{1}{2}$ m/s².

Now, we compute the next state, s_{k+1} , using only rational arithmetic:

$$\begin{aligned} p_{k+1} &= \frac{19}{2} + \left(\frac{3}{4}\right) \cdot \left(\frac{1}{5}\right) = \frac{19}{2} + \frac{3}{20} = \frac{190}{20} + \frac{3}{20} = \frac{193}{20} \\ v_{k+1} &= \frac{3}{4} + \left(-\frac{1}{2}\right) \cdot \left(\frac{1}{5}\right) = \frac{3}{4} - \frac{1}{10} = \frac{15}{20} - \frac{2}{20} = \frac{13}{20} \end{aligned}$$

5. **Verification:** The theorem prover must now check if the new state $s_{k+1} = (\frac{193}{20}, \frac{13}{20})$ is within the hypercube.
 - Check position: Is $0 \leq \frac{193}{20} \leq 10$? $193/20 = 9.65$. Yes, $0 \leq 9.65 \leq 10$.
 - Check velocity: Is $-1 \leq \frac{13}{20} \leq 1$? $13/20 = 0.65$. Yes, $-1 \leq 0.65 \leq 1$.

The state remains within the safe hypercube. A theorem prover would not just check one instance, but would symbolically prove that for *any* state (p_k, v_k) within the hypercube and for *any* valid control input, the resulting state (p_{k+1}, v_{k+1}) also lies in the hypercube. This single step serves as a concrete witness to the underlying arithmetic of such a proof.

B Appendix B: Counterexample Search for Invention 1

A claim's true strength is revealed not by confirmation, but by a failed search for refutation. Here, we probe for failure modes using SAT-style reasoning (i.e., can we satisfy the conditions for failure?).

1. **Failure Mode: Finite-Precision Overflow.** The claim specifies *finite-precision* rational arithmetic. A rational number $q = \frac{n}{d}$ requires storing a numerator n and a denominator d . Operations cause these to grow. For example, $\frac{n_1}{d_1} + \frac{n_2}{d_2} = \frac{n_1 d_2 + n_2 d_1}{d_1 d_2}$.

- **SAT-style Query:** Can we find a sequence of valid operations such that the numerator or denominator exceeds the maximum value storable in its finite-precision representation (e.g., a 64-bit integer)?
- **Counterexample Witness:** Consider a simple growth dynamic: $x_{k+1} = x_k \cdot \frac{5}{4}$. Let the initial state be $x_0 = 1$.

$$\begin{aligned} x_1 &= \frac{5}{4} \\ x_2 &= \frac{25}{16} \\ x_3 &= \frac{125}{64} \\ &\dots \\ x_k &= \frac{5^k}{4^k} \end{aligned}$$

If the finite precision is a 64-bit integer (max value $\approx 9 \times 10^{18}$), the numerator 5^k will overflow this limit when $k > \log_5(9 \times 10^{18})$, which is approximately $k = 28$. At the 28th time step, the system can no longer represent the state accurately, and the formal guarantee collapses. The claim’s mathematical purity fails an encounter with physical hardware limits.

2. **Failure Mode: Model-Reality Divergence.** The entire method proves properties of a *formal model*, not the physical system itself. The certificate guarantees the model’s safety, but the model is an imperfect abstraction of reality.

- **SAT-style Query:** Can there exist an unmodeled physical effect ϵ such that the true state $s_{true} = s_{model} + \epsilon$ is outside the hypercube, even when s_{model} is proven to be inside?
- **Counterexample Witness:** Let the model for our cart be $p_{k+1} = p_k + v_k \Delta t$. The theorem prover generates a certificate assuming this is true. Now, suppose the physical cart experiences unmodeled wheel slippage, sensor noise, or a 1% actuator error. If the model is at a state $p_k = 9.99$ (provably safe), a minor, unmodeled slippage of +0.02 meters would push the real cart to $p_{true} = 10.01$, outside the safe hypercube. The certificate remains valid, but the cart has crashed. The claim provides mathematical safety, but not necessarily physical safety.

3. **Failure Mode: Undecidability and Computational Complexity.** The claim implies the automated theorem prover can always generate a certificate for a safe system. For many classes of systems (e.g., non-linear, hybrid), the reachability problem is computationally undecidable.

- **SAT-style Query:** Does there exist a system with complex (e.g., non-linear) dynamics for which the theorem prover cannot terminate and thus cannot produce a certificate, even if the system is, in fact, safe?
- **Counterexample Witness:** Consider a system with dynamics involving trigonometric functions, e.g., $\theta_{k+1} = \theta_k + \omega \Delta t + \alpha \cos(\theta_k)$. Representing $\cos(\theta_k)$ in pure rational arithmetic requires a Taylor series approximation, which is itself a model inaccuracy. More profoundly, asking a theorem prover to find bounds on such a function over all possible inputs is a non-trivial, and in the general case, undecidable problem. The prover may run indefinitely, and per claim 1(g), the system would be forced into a safe shutdown, not because it was unsafe, but because its safety was too complex to be proven.

C Appendix C: Confidence Score for Invention 1

- **Mathematical Soundness Confidence Score: 65%**

Justification: The core idea of using deterministic, rational arithmetic to escape the non-associativity and representation errors of floating-point numbers is powerful and sound. It is a valid technique in formal methods. However, the claim is a significant overstatement of the practical reality. The score is not higher for three primary reasons discovered during our counterexample search:

1. It fails to address the inevitable growth and potential overflow of numerators/denominators in finite hardware, a critical failure mode.
2. It conflates safety of the model with safety of the physical apparatus, ignoring the unavoidable gap between a formal model and the real world.
3. It presumes computational tractability, whereas formal verification for any sufficiently complex (e.g., non-linear) system can be undecidable, meaning the prover may fail to produce a certificate for a perfectly safe system.

The claim is mathematically sound for a very restricted class of systems (e.g., simple linear systems where state growth is bounded and the model is near-perfect), but it is not a general solution for all cyber-physical systems as stated.

As requested, I will provide a formalization of the mathematical theorems underpinning the provided patent claims in Lean 4.

The request specifies three inventions; however, the provided document text details only one, titled “Invention 1: The Safety Kernel.” The text appears to be incomplete. I shall, therefore, provide a rigorous formalization for the single invention that was fully described. Upon receiving the specifications for the remaining two inventions, I will be prepared to formalize them with the same diligence.

Appendix B: Lean 4 Formalization

Invention 1: The Safety Kernel

The core mathematical assertion of this patent claim is that for a cyber-physical system whose dynamics can be modeled by functions with a specific structure—namely, using deterministic, finite-precision rational arithmetic (“Q-arithmetic”)—one can *computably generate* a formal, machine-checkable certificate of safety. This certificate provides a mathematical guarantee that the system’s state vector will remain within a predefined hypercube (the safe set) for a given time horizon.

The ability to generate this certificate automatically relies on the decidable theory of rational numbers. For a suitable class of functions (e.g., affine transformations), the problem of proving that a hypercube is an invariant set is decidable. This embodies the “Exact Rational Witness” pattern, where the certificate is not merely a boolean but a constructible proof term that the Lean 4 kernel can verify.

Below is the formalization. We model the “Q-arithmetic” dynamics as an affine transformation over rational numbers, which is a common and powerful enough model for many control systems.

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Of course. As a pragmatic engineer, I understand the critical importance of translating technical innovation into a viable business strategy. An invention, no matter how brilliant, only creates value when it solves a real-world problem and has a clear path to market.

I will now construct the requested business case for the first invention, "The Safety Kernel." Please provide the claims for the subsequent two inventions when you are ready, and I will perform the same analysis for them.

Here is the business case appendix for Invention 1.

D Appendix C: Business Case Analysis for Invention 1

D.1 TAM/SAM/SOM Market Sizing

The "Safety Kernel" technology addresses the market for safety-critical cyber-physical systems (CPS), where failure can lead to catastrophic outcomes.

- **Total Addressable Market (TAM):** The global market for Cyber-Physical Systems is our TAM. This market was valued at approximately USD 117.2 billion in 2023 and is projected to exceed USD 325 billion by 2030, growing at a CAGR of over 15%. This includes automotive, aerospace, industrial automation, and medical devices. (*Source: Grand View Research, "Cyber Physical Systems Market Size, Share & Trends Analysis Report"*).
- **Serviceable Addressable Market (SAM):** Our SAM is the segment of the CPS market that requires high-assurance formal verification for safety certification. This is primarily concentrated in autonomous vehicles (ISO 26262, SOTIF), aerospace (DO-178C), and industrial robotics (ISO 10218). We estimate this segment to be approximately 25% of the TAM, representing a serviceable market of **USD 29.3 billion** in 2023.
- **Serviceable Obtainable Market (SOM):** As a new entrant with disruptive technology, our initial goal is to capture a niche but high-value segment. We project capturing 1% of the SAM within the first 5 years, focusing on the autonomous automotive and industrial drone sectors. This represents a realistic target of approximately **USD 293 million** in annual revenue.

D.2 Competitive Landscape

The formal verification space is highly specialized. Competition comes from established tool providers and in-house development by large corporations.

- **Incumbents:**
 - **MathWorks:** Their Simulink and Polyspace products are deeply embedded in automotive and aerospace design workflows, offering model-based design and static code analysis. They represent the primary path of resistance in customer workflows.
 - **Ansys (formerly Esterel Technologies):** The SCADE product suite is a market leader for developing certified safety-critical embedded software, particularly in the aerospace sector under the DO-178C standard.
 - **EDA Giants (Synopsys, Cadence):** While focused on hardware verification, their formal verification engines are highly mature and could be adapted for software verification.
 - **Specialized Firms (e.g., AdaCore):** Companies like AdaCore provide specialized, high-assurance software language environments (SPARK Ada) that incorporate formal verification from the ground up.
- **Differentiation:** Our core differentiator is the guarantee of mathematical certainty by eliminating floating-point indeterminacy (via Q-arithmetic) and generating a machine-checkable certificate. This provides a level of assurance that current static analysis and model-checking tools cannot match, directly addressing the "black box" problem in AI-driven control systems.

D.3 Revenue Model

A multi-pronged revenue model is proposed to maximize market penetration and long-term value.

1. **Software Licensing (Core Revenue):** A tiered, per-seat annual license for our "Certus" design suite. Tiers would be based on the complexity of models and access to specialized toolboxes (e.g., Automotive, Aerospace). Expected ACV (Annual Contract Value) per customer: \$75k - \$250k.
2. **Verification-as-a-Service (VaaS) Platform:** A cloud-based SaaS platform where clients can upload system models for verification without high upfront software costs. This lowers the barrier to entry for smaller firms and is billed based on computational hours and model complexity.
3. **Professional Services & Support:** Integration support, certification consulting, and training services to help clients adapt their workflows and prepare for regulatory audits. This provides an upfront revenue stream and deepens customer relationships.

D.4 5-Year ROI Projection

The following projection is based on a successful Seed and Series A fundraise, enabling team growth and market entry. All figures are in USD thousands.

Metric	Year 1	Year 2	Year 3	Year 4	Year 5
Revenue					
Software Licenses	\$150	\$750	\$2,500	\$7,000	\$15,000
VaaS Platform	\$50	\$200	\$800	\$2,000	\$5,000
Professional Services	\$300	\$550	\$1,200	\$1,500	\$2,000
Total Revenue	\$500	\$1,500	\$4,500	\$10,500	\$22,000
Costs					
R&D	\$1,800	\$2,500	\$3,500	\$5,000	\$7,000
S,G&A	\$700	\$1,200	\$2,000	\$3,000	\$4,500
Total Costs	\$2,500	\$3,700	\$5,500	\$8,000	\$11,500
Net Profit/Loss	(\$2,000)	(\$2,200)	(\$1,000)	\$2,500	\$10,500
Cumulative P/L	(\$2,000)	(\$4,200)	(\$5,200)	(\$2,700)	\$7,800

Table 1: 5-Year Financial Projections (in USD thousands)

This projection shows profitability by Year 4, with a cumulative positive return achieved in Year 5, indicating a strong ROI for early-stage investors.

D.5 Fundraising Strategy

- **Seed Round:**
 - **Target:** \$3 Million
 - **Use of Funds:** Hire a core team of 8 (Formal Methods PhDs, Compilers, Systems Engineers), develop a minimum viable product (MVP) capable of verifying a benchmark system (e.g., an automated emergency braking system), and secure two paid pilot programs with automotive Tier 1 suppliers.
 - **Timeline:** Secure within 6 months.
- **Series A:**
 - **Target:** \$15 Million
 - **Prerequisite Milestones:** Successful completion of pilot programs, MVP demonstrating clear superiority over existing solutions, and a revenue pipeline of ~\$1M ARR.
 - **Use of Funds:** Scale the engineering team to 25, build a dedicated sales and marketing team, expand the product into the aerospace and defense vertical, and establish a European sales office.
 - **Timeline:** 18-24 months post-Seed.

D.6 Key Risks and Mitigations

- **Risk: Technical Scalability.** The state space of real-world systems (like a complete autonomous vehicle perception stack) can be too large for any theorem prover to handle ("state space explosion").
 - **Mitigation:** Our R&D roadmap prioritizes compositional verification, allowing for modules to be verified independently. We will initially focus on well-bounded subsystems (e.g., powertrain control, flight surface actuators) rather than entire systems.
- **Risk: Market Inertia.** Engineering workflows in target industries are extremely entrenched and resistant to change, with high training and tool-chain integration costs.
 - **Mitigation:** Develop "wrapper" plugins for existing environments like Simulink and Ansys SCADE. This allows engineers to use our verification engine on their existing models, dramatically lowering the barrier to adoption.
- **Risk: Talent Acquisition.** There is a very limited pool of engineers with expertise in automated theorem proving, compilers, and formal methods.
 - **Mitigation:** Forge partnerships with leading university labs (e.g., CMU, Stanford, Oxford). Sponsor research and establish a strong academic-to-industry pipeline. Offer compelling equity and the opportunity to work on foundational, industry-changing problems.

E Appendix D: Independent Peer Reviews

E.1 Peer Review 1: Reviewer Alpha

Model: gemini-2.5-pro **Verdict:** REVISE **Score:** 7/10

****TO**:** Xavier Callens, Socrate AI Lab ****FROM**:** Reviewer Alpha, USPTO ****DATE**:** 2026-06-14 ****SUBJECT**:** Independent Peer Review of Patent Portfolio ("Novel Safety Verification Systems")

This document constitutes a formal review of the patent application for "Invention 1: The Safety Kernel." My analysis is based on the provided claims, numeric validation, and Lean 4 formalization. The review of subsequent inventions is pending receipt of their documentation.

1. VERDICT: ****REVISE****

The application presents a novel and intellectually interesting approach to a critical problem in safety engineering. However, the claims as written are overly broad and suffer from a significant lack of enablement, rendering them untenable in their current form. The core concept may be patentable, but it requires substantial revision to narrow the scope and adequately describe the implementation of its most critical steps.

2. PATENTABILITY SCORE: ****4/10****

The score reflects a promising, novel synthesis of ideas (Q-arithmetic + theorem proving for predictive safety) that is severely undermined by abstract claiming and a failure to enable the invention across the proposed scope. The gap between the trivial 1D example and the claim's applicability to general "cyber-physical systems" is too wide.

3. STRENGTHS

****Addresses a Foundational Problem:**** The invention directly confronts the issue of floating-point indeterminacy, a well-known and dangerous source of errors in safety-critical computation. Proposing a solution that formally eliminates this entire class of errors is a significant strength. ****High Assurance via Formal Proof:**** The generation of a machine-checkable certificate is the gold standard for high-assurance systems. It provides a level of certainty that is mathematically superior to traditional simulation, testing, or model-checking with bounded precision. ****Novel Synthesis of Techniques:**** While formal methods, theorem provers, and rational

E.2 Peer Review 2: Reviewer Beta

Model: gemini-2.5-flash **Verdict:** REVISE **Score:** 7/10

I cannot provide a complete review of the patent portfolio as the provided patent claims are incomplete. Specifically, the "System Claim" for Invention 1 is truncated, and the details for Inventions 2 and 3 are entirely missing.

To offer a thorough and accurate peer review, I require the complete text of all claims, including the full descriptions for all three inventions, as well as the complete "Numeric Validation" and "Lean 4 Formalization" appendices.

Based on the **"Invention 1: The Safety Kernel - Method Claim"** provided:

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PEER REVIEW: Preliminary Assessment for Invention 1 (Method Claim only)

1. **VERDICT:** REVISE (Pending full claims) 2. **PATENTABILITY SCORE:** 7/10 (Based solely on the method claim. Subject to change with full information and prior art search.)

3. **STRENGTHS:** **Novelty in Q-arithmetic for Safety:** The explicit use of "deterministic, finite-precision rational number arithmetic (Q-arithmetic)" to eliminate floating-point indeterminacy during formal verification is a strong and potentially novel approach to a well-known problem in safety-critical systems. This addresses a fundamental source of imprecision that current methods often struggle with. **Machine-Checkable Certificates:** The generation of "machine-checkable certificates" provides a robust and independently verifiable proof of safety, which is highly valuable in regulated industries requiring stringent safety standards and audit trails. **Formal and Rigorous Approach:** The method outlines a clear, formal verification pipeline, moving from state definition and safe bounds to dynamic modeling, Q-arithmetic translation, automated theorem proving, and certified output. This structured approach underpins strong defensibility. **Clear Safety Response:** The inclusion of "corrective action or a safe shutdown protocol" if a certificate cannot be generated shows a practical, safety-first design.

4. **WEAKNESSES:** **Computation**

E.3 Peer Review 3: Reviewer Gamma

Model: gemini-2.0-flash **Verdict:** REVISE **Score:** 7/10